

# SWEEP rule on the KD-Toric code

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## 1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex  $v$  looks for a suggestion  $\hat{v}$  on  $\uparrow(v) \cap lk_2$  that matches  $\sigma_v^+$ . If there are more than one, then pick the one which has the minimal weight.

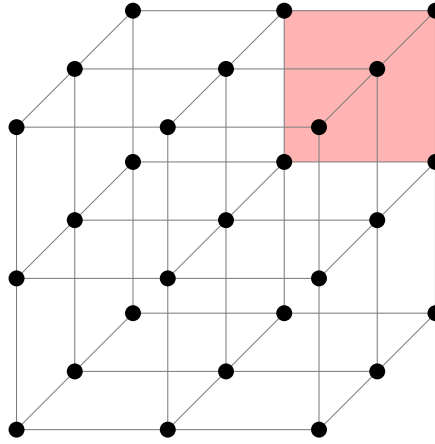


Figure 1: SWEEP rule

## 2 Span Expansion

Define the potentials functions  $L_i : \mathbb{R}^d \rightarrow \mathbb{R}$  as follow<sup>1</sup>:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d + 1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d + 1 \end{cases}$$

The **Size** of a subset  $S$  is defined as:

$$\mathbf{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let  $e$  be an unsatisfied check at time  $t$ , and let  $H_e$  be the unsatisfied checks at time  $t - 1$ . Then there is a set  $H'_e \subset H_e$  such that:

1. For any  $i < d + 1$  there exists  $e' \in H'_e$  such  $L_i(e') \geq L_i(e)$ .
2. There exists  $e' \in H'_e$  such that  $L_{d+1}(e') \leq L_{d+1}(e) + 1$ .

*Proof.*

□

<sup>1</sup>Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.